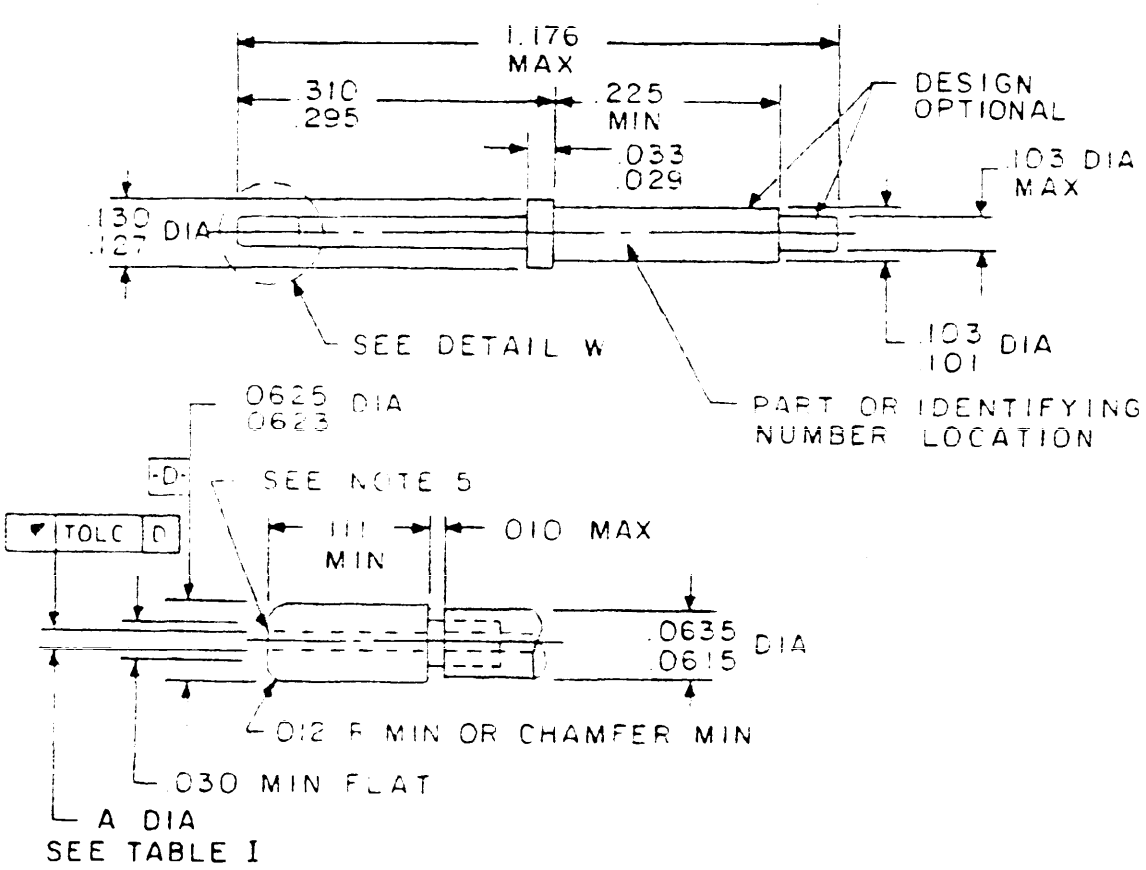
MIL-T-29504/4B
7 June 1991
SUPERSEDING
MIL-T-29504/4A
23 April 1990

MILITARY SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, CLASS 5, TYPE II, STYLE A, PIN TERMINUS, SIZE 16, REAR RELEASE MIL-C-38999, SERIES I, III, and IV

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-29504.



DETAIL W

FIGURE 1. Pin terminus.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm		Inches	mm
.002	0.05		.101	2.57
.012	0.30		.103	2.62
.029	0.74		.125	3.18
.030	0.76		.127	3.23
.033	0.84		.130	3.30
.0575	1.461		.225	5.72
.0615	1.56		.295	7.49
.0623	1.582		.310	7.87
.0625	1.588		.525	13.39
.084	2.13		1.176	29.87

NOTES:
1. Dimensions are in inches, except diameter "A" which is dimensioned in metric microns only.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .003 unless otherwise indicated.
4. Dimensions shown after plating.
5. Polished fiber face may be flat to concave and flush to recess.
6. "A" diameter hole tolerance is +8 -0. See table I.

FIGURE 1. Pin terminus - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Weight: 1 gram maximum.

Adhesives: Use a B-stage epoxy with a curing time of 15 minutes at 200°C. Epotec 354, or Epotec 353ND, or an equivalent approved by the preparing activity shall be used. Curing instructions shall be supplied with the assembly instructions provided with each termini or applicable connector.

Tools: See table II.

Mating terminus: MIL-T-29504/5.

Part or Identifying Number (PIN): See figure 1 and table I. The PIN M29504/4- (dash number from table I.)

Marking: The PIN shall be marked on barrel of the termini.

Temperature life: The upper temperature limit shall be 150°C ±3°C.

Thermal shock: The temperature range used shall be -65°C to +150°C.

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TABLE I. "A" diameter hole for optical waveguide fiber and PIN designation.

TICC	PIN	"A" diameter 1/	Total allowable offset tolerance 2/		Fiber size	Terminus
		μ	Circular runout 3/	Max $\frac{B + C}{2}$ (inch)μ 4/	Core/clad μ	Size
4039	M29504/04-4039	124.5 +B -0	C			16
4040	M29504/04-4040	127.0 +B -0	C	(.00059) 15	50/125	16
4041	M29504/04-4041	129.0 +B -0	C			16
4042	M29504/04-4042	139.5 +B -0	C			16
4043	M29504/04-4043	142.0 +B -0	C	(.00059) 15	100/140	16
4044	M29504/04-4044	144.0 +B -0	C			16

4/ Dimensions in parentheses are inches.
3/ As defined by ANSI Y14.5M - 1982.
1/ "B" is defined as the clearance between the fiber and the hole. B and C are not specified, but calculated or selected by design.
2/ The total allowable offset tolerance is the tolerance that controls the offset of the fiber to the alignment diameter of .0625/.0623.

TABLE II. Tools.

End size	Insertion tool	Removal tool
16	M81969/14-03	M81969/14-03

Patent notice. The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
3,947,182	3/30/1993
4,351,586	9/28/1997
4,747,658	5/31/2005

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 17, 19, 70, 80, 90, 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC, OS, YD
Air Force - 13, 14, 84

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0092-1)

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